00713



Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination
Year 2, Semester 2 (2022)

IT2020 – Software Engineering

Duration: 3 Hours

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Instructions to Candidates:

- This paper has 5 questions.
- Answer all questions in the booklet given.
- The total mark for the paper is 100.
- This paper contains 10 pages, including the cover page.
- Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 1 (25 Marks)

Read the scenario and draw a **sequence diagram** using suitable boundary, control, and entity classes. (The class structure is given below)

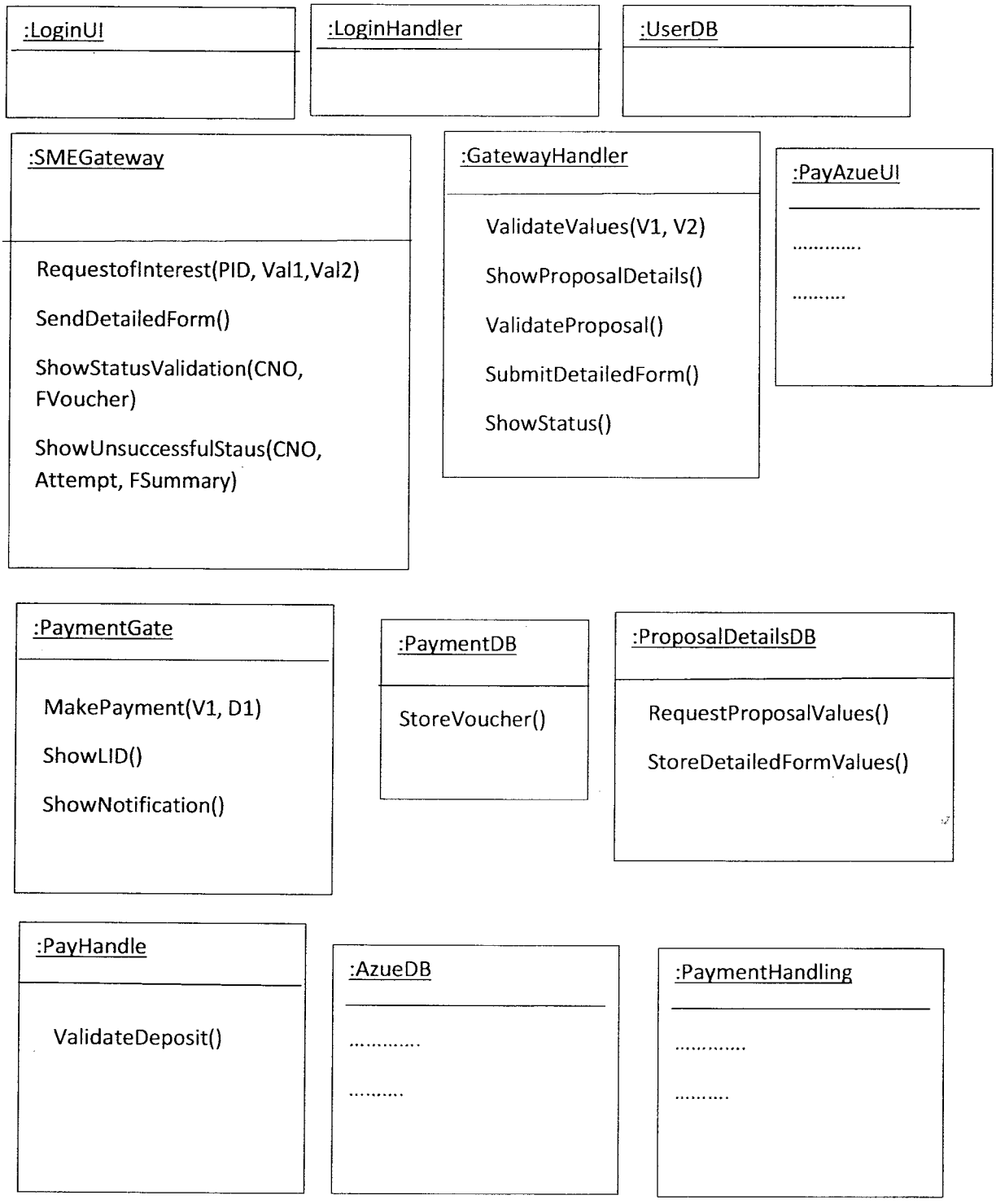
A project called " SRILAKA" assists the Sri Lankan community's Small and Medium-sized growing Enterprises (SMEs). Moreover, "SRILAKA" offers SME connect platforms that offer them access to a wealth of information and expertise. Also, it provides many services to the community who register with them. Due to the numerous advantages, many consumers are drawn to the SME licensing and registering service.

Initially, all the entrepreneurs need to log in to the system to start the process. To license any of the SMEs, the entrepreneurs must send a request with the proposal. This proposal should be properly made with the values that are already instructed on the client's information website. So, the proposal will be validated according to the predetermined values. These predetermined values are stored in ProposalDetailsDB to access the validation process. At the same time, the system sends the detailed form to the user for further details

Then the entrepreneur may submit the detailed form to the system and the values will be stored in the ProposalDetailsDB. So, a successful status of storing the details will be triggered by the system indicating the status of the validation process. Further, the system shows the Company No and financial voucher to proceed with the payment. If the submitted proposal is in an unacceptable range, the system will show the Company No, attempt, and failure summary to the user. So, entrepreneurs must keep track of this Company No for further processing.

An entrepreneur must submit the financial voucher with deposit proof as instructed on the website. The system will store the voucher for further processing. Concurrently, the system will forward the proof of deposit to a third-party financial body to access the validity. Unless the status of the third-party financial body is unsuccessful, the system will show the License ID, document list, and the sample labeling method (if applicable) to the user. Moreover, to obtain the license, the entrepreneur must submit the documents with the license ID physically within one month of obtaining the license ID through the system. If a third-party financial body's status is unsuccessful, the system must show insufficient funds notification with the proper ID. For one entrepreneur, there can be many SME registrations to obtain the licensing.

Note: Please note that login to the system and check deposit through third-party company are separately modeled as "Login" and "Validate Deposit" respectively.


:LoginUI
:LoginHandler
:UserDB
:SMEGateway
RequestofInterest(PID, Val1,Val2)
SendDetailedForm()
ShowStatusValidation(CNO, FVoucher)
ShowUnsuccessfulStaus(CNO, Attempt, FSummary)
:GatewayHandler
ValidateValues(V1, V2)
ShowProposalDetails()
ValidateProposal()
SubmitDetailedForm()
ShowStatus()
:PayAzueUI
.....
.....
:PaymentGate
MakePayment(V1, D1)
ShowLID()
ShowNotification()
:PaymentDB
StoreVoucher()
:ProposalDetailsDB
RequestProposalValues()
StoreDetailedFormValues()
:PayHandle
ValidateDeposit()
:AzueDB
.....
.....
:PaymentHandling
.....
.....

Question 2 (20 Marks)

Given below is a detailed description of a web-based Vehicle Registration and Licensing application developed for the Department of Motor Traffic. Model a **physical diagram** for the following description and answer the questions given below.

Note: You MUST use appropriate operating systems and ports in the required places.

This application can be used by both mobile and desktop/laptop users to register their vehicles or to reserve a date for the written exam to obtain the driving license. Desktop users access this web application via DMT_webClient application through the web browser while Mobile users access this web application through DMT_MobApp application.

The Vehicle_Registry application resides inside Sun Java System Web Server which is deployed in a Lenovo physical server. The Vehicle_Registry application contains two sub-components: Registration and Ownership. Both Desktop and Mobile devices access Registration and Ownership components through iRegistration and iOwnership interfaces respectively.

An exam candidate can register to the system through a PC or mobile phone by accessing the i_Candidate interface of Candidate_Registration_UI, developed using JSP web pages. It resides inside IIS Webserver of Dell physical server.

The main Vehicle_Management application which contains three sub-components; Authentication, vehicleDB and License are installed in a separate application server that runs Oracle_Solaris_10 operating system of an Oracle x86_Server. I_Authentication interface realized by the Authentication component is used by Registration and Candidate_Registration_UI components. I_VehicleDB interface realized by the vehicleDB component is used by License component through its I_VehicleLisence interface. The Vehicle_Management component uses Ownership component through I_owner interface. User.jar file which resides in Vehicle_Management application manifests the Authentication component.

Both Sun Java System and IIS servers are connected to desktop and mobile devices via http protocol. Oracle x86_Server is connected to Lenovo ThinkServer and Dell Hardware server via a wide area network.

Question 3 (20 Marks)

Part 01

Based on the given code, come up with the answers to the below questions. (11 marks)

```
#include<stdio.h>
int main()
{
   int num = 0, even = 0, odd = 0;
   do {
       if( num % 2 == 0)
        {
            printf( "%d is an even number \n", num );
            even++ ;
        }
        else
        {
            printf( "%d is an odd number\n", num );
            odd++ ;
        }
        num++ ;
   } while ( num <= 10) ;
   printf( "Even count : %d\n" , even ) ;
   printf( "Odd Count : %d\n" , odd );
}
```

a) Draw the control flow graph for the given code. (03 marks)

b) How many branches are there in the given code segment? (01 mark)

c) Calculate the minimum number of test cases required for full statement coverage. (Show the statement coverage as a percentage for each test case). (03 marks)

d) When num=11, calculate the branch coverage percentage. (02 marks)

e) When num=9, calculate the branch coverage percentage. (02 marks)

Part 02

Based on the given code, come up with the answers to the below questions. (09 marks)

```
#include<stdio.h>
int main()
{
    char type ;
    float balance = 0 , amount ;
    printf( "Enter the transaction type :( w - Withdrawal d -Deposit) : " );
    scanf( "%c" , &type ) ;
    printf( "Enter the balance : " );
    scanf( "%f" , &balance ) ;
   switch(type)
    {
        case 'w' : printf( "Enter the withdraw amount : " ) ;
           scanf( "%f" , &amount ) ;
           if (balance > amount )
                balance = balance - amount ;
           break ;
        case 'd' : printf( "Enter the deposit amount : " ) ;
           scanf( "%f" , &amount );
           balance = balance + amount ;
           break ;
       default : printf( "Invalid Transaction\n" ) ;
            balance = balance ;
    }
   printf( "Current balance : %.2f\n" , balance );
}
```

a) Draw the control flow graph for the given code. (02 marks)

b) How many branches are there in the given code segment? (01 mark)

c) Calculate the minimum number of test cases required for full statement coverage (Show the statement coverage as a percentage for each test case). (06 marks)

Question 4 (20 marks)

a) The *meteorology* department recently launched a new online weather system to capture weather data. The Weather data class measures the weather using three parameters: humidity, pressure, and temperature. This information should be available to *meteorology* staff to display, and there are three types of displays namely, current, statistics, and forecast. The current display shows the current weather conditions, the statistic display represents statistics for the given period and the forecast display represents weather conditions for the next three days.

i. What design pattern will you use when modeling this scenario? State your reason(s). (02 marks)

ii. Draw the class diagram for the identified design pattern with relevant to the scenario. (05 marks)

b) A customer in a restaurant orders food from the menu, which is probably described in half a line. The order goes to the kitchen and the food will be served after a while. The customer doesn't want to know who prepares the ingredients, how long and how it will be cooked, and who is going to serve it afterwards. The customer just wants to eat a tasty meal that meets his expectations.

i. What is the design pattern that you will use when modeling this scenario? State your reason(s). (02 marks)

ii. Draw the class diagram for the identified design pattern with relevant to the scenario. (03 marks)

c) Consider the following code segment and answer the subsequent questions.

```
public interface Food {
    public String prepareFood();
}

public class VegFood implements Food {
    public String prepareFood(){
        return "Veg Food";
    }
}
```

```
public abstract class FoodType implements Food{
    private Food newFood;
    public FoodType(Food newFood){
        this.newFood=newFood;
    }

    @Override
    public String prepareFood(){
        return newFood.prepareFood();
    }
}

public class VegFood extends FoodType{
    public VegFood(Food newFood){
        super(newFood);
    }

    @Override
    public String prepareFood(){
        return super.prepareFood() +" With Rice and Vegetables  ";
    }
}

public class NonVegFood extends FoodType{
    public NonVegFood(Food newFood){
        super(newFood);
    }

    @Override
    public String prepareFood(){
        return super.prepareFood() +" With Roasted Chiken and Chiken Curry  ";
    }
}

public class ChineeseFood extends FoodType {
    public ChineeseFood(Food newFood){
        super(newFood);
    }

    @Override
    public String prepareFood(){
        return super.prepareFood() +" With Fried Rice and Manchurian  ";
    }
}
```

```
public class Customer {
    private static int  choice;
    public static void main(String args[]){
        do{
            System.out.print("========= Food Menu ============ \n");
            System.out.print("            1. Vegetarian Food.   \n");
            System.out.print("            2. Non-Vegetarian Food.\n");
            System.out.print("            3. Chineese Food.         \n");
            System.out.print("            4. Exit                        \n");
            System.out.print("Enter your choice: ");
            BufferedReader br=new BufferedReader(new InputStreamReader(System.in));

            choice=Integer.parseInt(br.readLine());
            switch (choice) {
                case 1:{
                  VegFood vf=new VegFood();
                  System.out.println(vf.prepareFood());
               }
               break;

               case 2:{
                   Food f1=new NonVegFood((Food) new VegFood());
                   System.out.println(f1.prepareFood());
               }
               break;
               case 3:{
                   Food f2=new ChineeseFood((Food) new VegFood());
                   System.out.println(f2.prepareFood());
               }
               break;

               default:{
                   System.out.println("Other than these no food available");
               }
               return;
           }

       }while(choice!=4);
   }
}
```

i. Identify the design pattern that has been implemented in the above code segment. (01 mark)

ii. What is the main purpose of using the design pattern you have mentioned above for this solution? (02 marks)

iii. Draw the class structure of the design pattern that you identified in part (i) with appropriate classes and methods according to the above scenario. (05 marks)

Question 5 (15 marks)

a) With the increased usage of smartphones over the period of the pandemic, more software companies moved into the development of mobile applications to provide entertainment, education, banking services, and so on. A well-known virtual reality game development company decided to expand its business further with more advanced games among teenagers with the popularity of mobile games. Initially, they came up with a version with basic features and planned to further improve based on clients' feedback and ratings.

i) Explain how Version control management supports the above business approach. Give reasons for your explanations. (04 marks)

ii) Explain the process of handling change requests after the initial release. (03 marks)

b) An object state is a status of an object at a given time. States have different internal behaviors. State different internal behaviors of a state and describe them with suitable examples. (04 marks)

c) Triggers make objects change their state from one state to another. What are the three different triggers in a state chart diagram? Explain with suitable examples. (04 marks)

---------- END OF THE QUESTION PAPER ----------